

AI Detection Service for Phone Interviews

Project Overview

The goal of this project is to develop an **API service** (and optionally a Python library) that can detect whether spoken or written responses during interviews are **AI-generated, scripted, or human-originated**. This system will support our broader platform by integrating seamlessly with both **post-call analysis pipelines** and **real-time AI-texting services**, providing actionable confidence scores on the authenticity of candidate responses.

Key Features

- **API-first Design**
 - Accepts either:
 - **Audio blob** (e.g., `.wav`, `.mp3`)
 - **Text string** (transcribed response)

Returns:

```
{  
  "confidence_score": 0.87,  
  "label": "AI-generated"  
}
```

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- **Multi-Modal Input**
 - Supports raw **audio analysis** (prosody, cadence, pauses).
 - Supports **text-based analysis** (linguistic features, perplexity, burstiness).
- **Scalable & Modular**
 - Exposed as a microservice with REST API endpoints.

- Can be containerized (Docker) for deployment.
 - Designed for integration with other applications.
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Proposed Technical Approach

1. Text Detection

- Fine-tune or adapt existing models (e.g., OpenAI's detector, GPTZero, DetectGPT-style methods).
- Use metrics like **perplexity, burstiness, repetition patterns**.

2. Audio Detection

- Extract **speech features** (MFCCs, pitch, pauses, prosody).
- Train classifier on real human audio vs. AI TTS audio.

3. Service Layer

- Build an API (Flask/FastAPI) with endpoints:
 - `/detect/text` → input string
 - `/detect/audio` → input audio blob
- Return JSON results with **confidence scores**.

4. Deployment & Integration

- Package as a **standalone service** deployable on Docker.
 - Provide client SDK (Python, JS) for ease of integration into existing systems.
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Example Use Cases

- **Post-call analysis:** Detect if an interviewee's responses were scripted or AI-assisted.
 - **AI-texting pipeline:** Screen for AI-generated candidate replies during asynchronous assessments.
 - **Research & benchmarking:** Provide data on how often AI assistance is used in recruiting workflows.
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Next Steps

1. Define **scope for MVP** (text-only vs. text+audio).
 2. Research and evaluate **baseline detectors** (e.g., GPTZero, DetectGPT, OpenAI API).
 3. Set up **API skeleton** (FastAPI + Docker).
 4. Prepare **datasets** for fine-tuning & evaluation.
 5. Iteratively develop & benchmark models.
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Resources

- [DetectGPT Paper](#)
- [OpenAI AI Text Classifier](#)
- [GPTZero](#)
- [Librosa \(Audio Analysis Library\)](#)
- [FastAPI Framework](#)